

Deep diving into the Chronicles of Robotic Process Automation RPA

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You just came back from a work trip. You're exhausted (zero stamina even after all that gymming). You come back with 12 crumpled receipts, including one mysterious coffee charge (you don't even remember drinking coffee) and a vague memory of 'networking' over some sushi.

Now comes the real pain: filling in the expense report.

- Open excel
- Copy dates from the receipt
- Enter the amount
- Pick the category (think deeply, was the sushi 'meals' or 'client entertainment'?)
- Upload receipt pdf
- Repeat for all the other 11 receipts.

By the time you're done, you don't want reimbursement from your office, you crave retirement (with some good pension amount).

Now, let us rewind the imagination and add an impact player in the process. An RPA bot.

- It reads your receipts (even with ketchup stains)
- Matches them to your calendar and card statements
- Fills out the form
- Submits the form for approval even before you've even unpacked

Expense report done while you're looking for your missing sock or wondering when you drank that coffee.

This is what Robotic Process Automation (RPA) is. It saves time and sanity.



What Is RPA?

Robotic Process Automation (RPA) is a technology introduced in the 1980s, that enables businesses to automate repetitive and rule-based tasks by configuring software robots that mimic human actions. These bots are capable of interacting with digital systems and applications just like humans do and very well a master in performing tasks such as data extraction, entry, form filling and transaction processing.

Main Consumption:

RPA endeavours to enhance operational efficiency, precision, and productivity by automating routine tasks, thereby liberating human workers from mundane activities and enabling them to concentrate on higher-value tasks.

Understanding RPA:

RPA is a sophisticated software technology that automates rule-based, repetitive tasks by replicating human actions within applications and systems. It functions as a digital workforce, executing processes with precision and consistency, without the limitations of fatigue or distractions.

The Strategic Impact of RPA Across Industries:

RPA has demonstrated significant value in various sectors, optimizing business operations and driving productivity. Some key areas of impact include:

- **Finance & Accounting:** Automated invoice processing and financial reconciliations.
- **Healthcare:** Improved patient data management, insurance claims processing.
- **Customer Service:** Enhanced service delivery through automated query responses.
- **Human Resources:** Efficient recruitment processes, employee onboarding, and payroll automation.

Organizations leveraging RPA experience improved operational efficiency, cost savings, and higher accuracy in execution.



Addressing Concerns: RPA as an Enabler, not a Replacement:

A common misconception is that RPA is designed to replace human workers. It serves as an enabler, allowing employees to focus on higher-value tasks such as strategic decision-making, problem-solving, and innovation. By automating routine activities, RPA empowers professionals to engage in more meaningful and complex work.

Embracing RPA for Competitive Advantage:

For businesses still relying on manual processes, the adoption of RPA presents a significant opportunity for transformation. Whether optimizing workflows, improving customer experiences, or reducing operational costs, RPA provides a pathway to greater efficiency and innovation.

The era of digital automation is here, and organizations that embrace it will be well-positioned for future success.

Let us dive into a hands-on example!

Intelligent Document Processing through RPA:

I have created this very simple invoice processing bot to showcase the application of RPA. It streamlines a very common business operation- invoice data entry.

I have used UiPath to develop a bot that mimics human actions to extract invoice information from PDF files and logs it into an excel sheet, eliminating manual and repetitive work.

Technologies and Tools Used:

UiPath Community Studio: A low-code automation platform, that offers drag and drop interface of pre-built components.

Microsoft Excel: For the output logs.

What the RPA bot does:

1. Monitors a folder: watches a directory for new invoice PDFs to process
2. Reads and parses PDF invoices: extracts key invoice fields like invoice number, invoice date, vendor name and amount due.
 - We can add any number of fields and specify a regex to identify the match while parsing
3. Validates extracted data: ensures all fields are present and correctly formatted.
4. Writes to excel log: appends the structured data to an excel sheet.

Automation steps in brief:

1. **Set up folder paths and sample invoices**
Create the folder structure (Invoices\New, Output, and optionally Logs) and add sample invoice PDFs.
2. **Create a UiPath workflow with a main Sequence**
Design the automation using a Sequence container and initialize required variables.
3. **Loop through PDFs using 'For Each File in Folder'**
Iterate over each file in the invoice folder.
4. **Use 'Read PDF Text' and Regex to extract values**

5. Extract invoice number, date, vendor, and amount using regular expressions.
6. **Use 'Add Data Row' to add info to a DataTable**
Add the extracted values into a row of a DataTable object.
7. **Use 'Append Range' to write to an Excel file**
Log the DataTable content into an Excel file (invoice_log.xlsx).

RPA Benefits Demonstrated:

1. Saves time: eliminates manual invoice entry
2. Accuracy: minimizes human error using regex-based extraction
3. Scalability: Can handle hundreds of invoices at once
4. Auditability: Output logs and excel records acts as documentation
5. Low Code/No Code: Achieved through UiPath Studio's visual interface

Challenges and limitations:

1. Requires consistent invoice formatting, for example if we set the regex to identify "Invoice Number" and not "Invoice No", we lose out on information.
2. Regex must be tuned to different layouts.

The Future of RPA: Intelligent Automation

As RPA continues to evolve, its integration with Artificial Intelligence (AI) and Machine Learning (ML) is ushering in a new era of Intelligent Automation. These advancements enable RPA solutions to not only follow predefined rules but also learn, adapt, and make informed decisions. This shift enhances automation capabilities, making business processes more responsive and intelligent.

Key Takeaway:

RPA bots can take over tedious, rule-based, high-volume tasks like invoice entry — saving time, reducing errors, and improving operational efficiency.

Here's the YouTube link to the invoice processing bot demonstration:

<https://youtu.be/RpurKfBdWjY>